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Video course
ML, DL, NLP
Fresh Noise Sir

Sec-22 Probability
Distribution Function
for Data

lec-5 Poisson Distribution

lec-6 Normal Gaussian
Distribution

lec-7 Standard Normal
Distribution

In Probability theory
& statistics, the
Poisson distribution is
a discrete probability
distribution that expresses
the probability of a
given number of events
occurring in fixed intervals
of time.

It is, if there events
occur with a

(1) Discrete Random variable
(P.M.F)

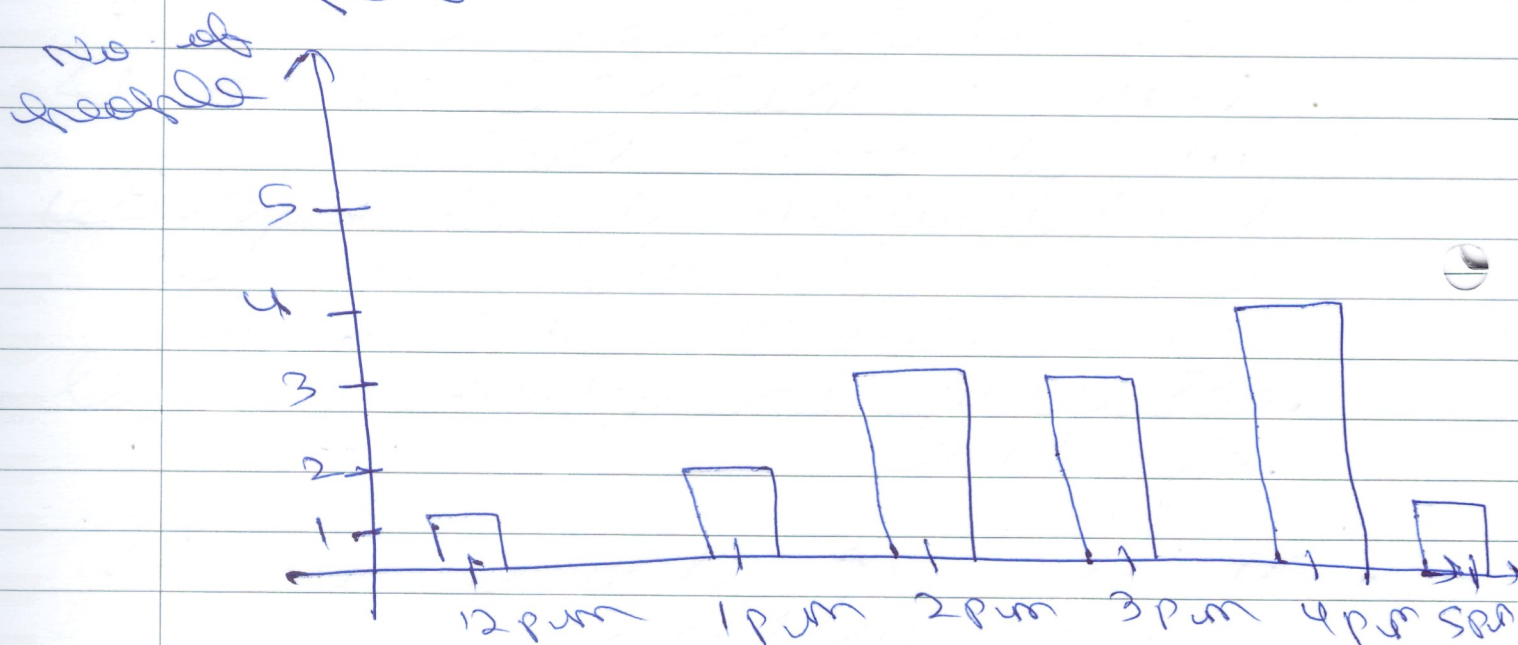
P.T.U

(2) Describe the number of events occurring in a fixed time interval.

eg:- No. of people visiting hospital every hour.

No. of people visiting bank every hour.

(*) The graph can be plotted as:-



$\lambda = 3 \Rightarrow$ expected no. of events occurring at average time interval

(3)

(*) Now, the PMF used over hour is :-

Q:- The probability of the person visiting at 3rd hour?

Soln.

In PMF,
 $P(X=3) \Rightarrow$

$$\frac{e^{-\lambda} \lambda^x}{x!}$$
$$= \frac{e^{-3} \times 3^3}{3!}$$

$$= 0.101$$

$$= 10.1\%$$

probability

This is the required probability

(*) Now we have to calculate the probability of person visiting at 4th hour is?

In PMF,

$$P(X=4) \Rightarrow$$

$$P(X=4) + P(X=3)$$

$\Rightarrow$ Find Probability

mean of poisson distribution

$$\text{mean} = E(x) = \mu = \lambda \Delta t$$

Variance

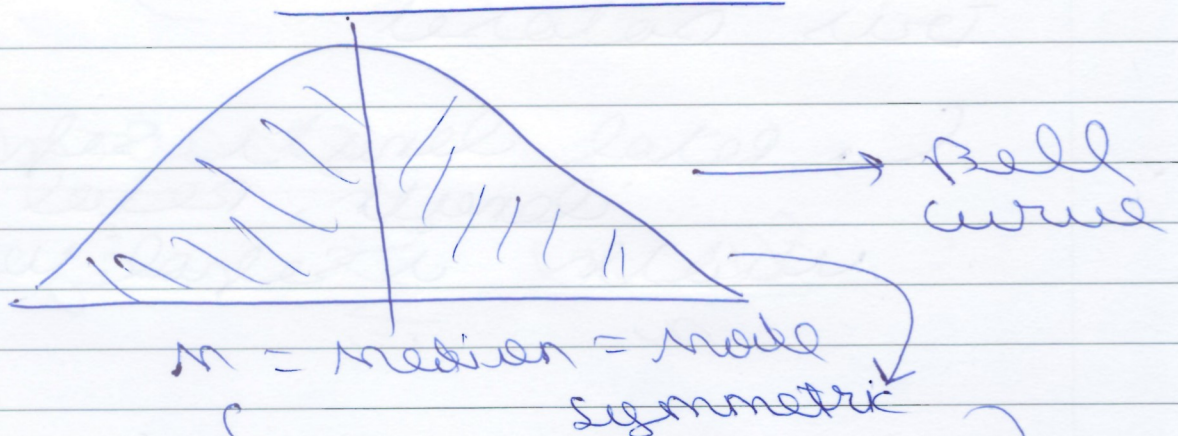
λ = Expected No. of ~~the~~ events occur at every time interval

Δt = time interval

lec-6Normal / Gaussian distribution

In probability theory and statistics, a normal distribution or Gaussian distribution is a type of continuous probability distribution for a real valued random variable.

① Continuous Random Variable :-



$X = \{ \text{---} \}$

→ this is how continuous random variable looks like.

Notation :- $N(\mu, \sigma^2)$

Parameters :- $\mu \in \mathbb{R} = \text{mean}$
 $\sigma^2 \in \mathbb{R} > 0 =$
variance

Example: let's consider
weights of
students in a class.

Also the heights
of students in a class

One of the most
common dataset is
Iris dataset.

→ Petal length, Sepal
length, Petal
width & Sepal width

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For a Normal/Gaussian Distribution, we follow the PDF.

$$PDF = \frac{1}{\sigma \sqrt{2\pi}} e^{-1/2 \left(\frac{x_i - \mu}{\sigma} \right)^2}$$

↙
This is how PDF function is created

① Mean of Normal Gaussian

$$\mu = \frac{\sum_{i=1}^n x_i}{n}$$

② Variance

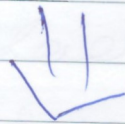
$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$$

③ Standard Deviation
 $\sigma = \sqrt{\text{variance}}$

Empirical Rule of Normal / Gaussian Distribution

Suppose we have a random variable, where we have 50 data points.

$$X = \{ \text{---} \}$$



Normal / Gaussian Distribution

These assumptions are related to probability.

$$\Pr(\mu - \sigma \leq X \leq \mu + \sigma) \approx 68\%$$

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 95\%$$

$$\Pr(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 99.7\%$$

lec-7

①

Standard Normal Distribution & Z-Score

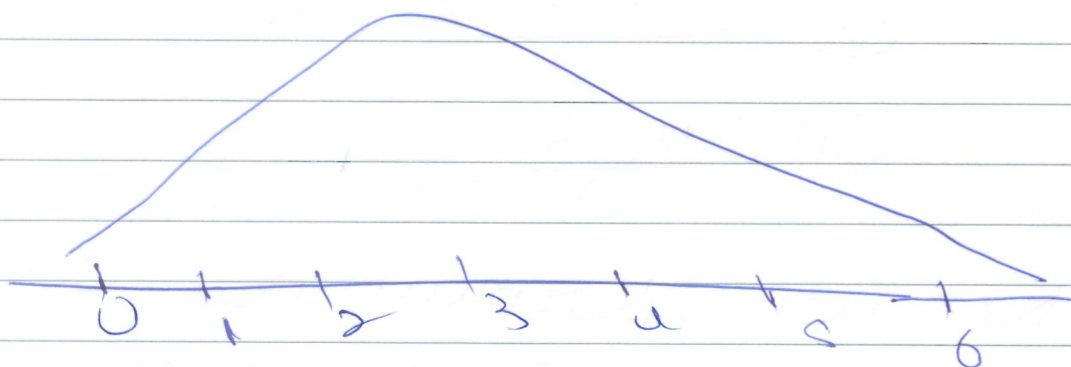
Suppose we have a random variable which has the values like as :-

$$X = \{1, 2, 3, 4, 5\}$$

$$\mu = 3 \quad (\text{mean})$$

$$\sigma = \sqrt{1} \approx 1 \quad (\text{standard deviation})$$

Bell curve here is as:-



Random Variable

$$X = \{1, 2, 3, 4, 5\}$$

Mean = 0 & standard deviation = 1

One type of distribution where we get $\mu = 0$ & $\sigma = 1$

→ Then that distribution is known as Normal standard distribution

$$z\text{-score} = \frac{x_i - \mu}{\sigma}$$

$$y = \{-2, -1, 0, 1, 2\}$$

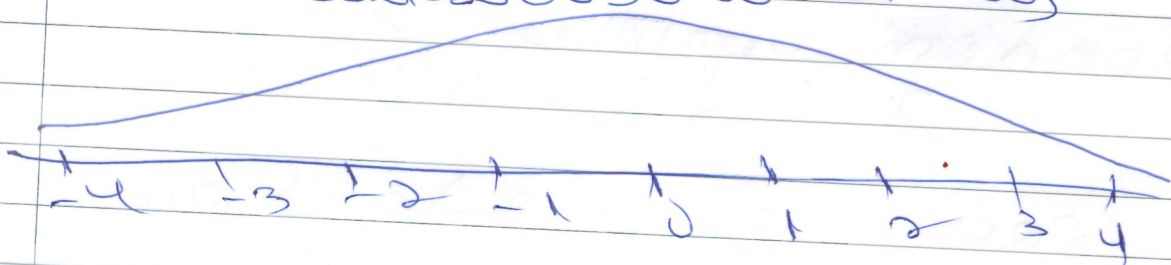
$$(1) \frac{1-3}{1} = -2$$

$$(2) \frac{2-3}{1} = -1$$

$$(3) \frac{3-3}{1} = 0$$

$$(4) \frac{4-3}{1} = 1$$

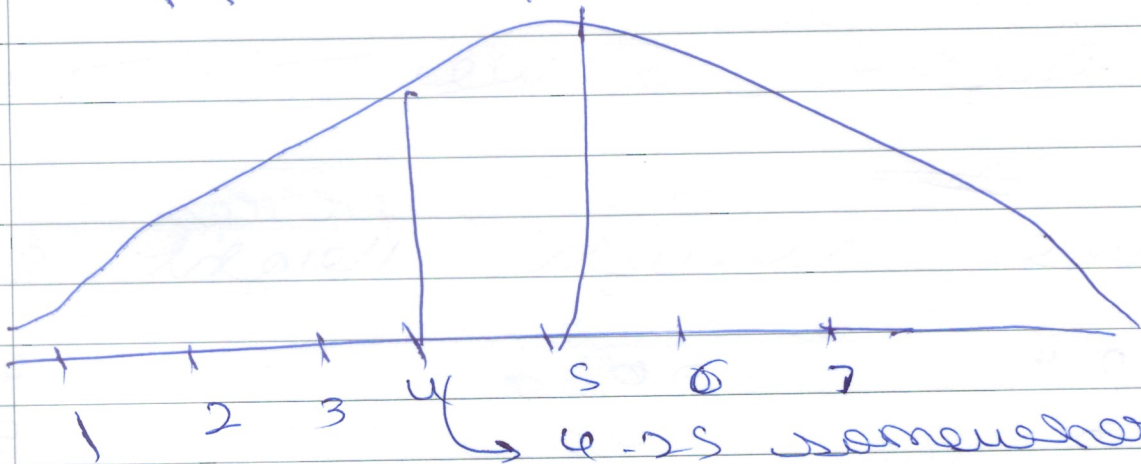
Now drawing the distribution



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$X \approx$ Standard Normal
Distribution
($\mu = 0, \sigma = 1$)

Suppose $\mu = 4, \sigma = 1$



Q: How many standard
deviations 4.25 is away
from mean?

Use 4.25 as x or 4.25

$$x_i = 4.25$$

$$z \text{ score} = \frac{4.25 - \overset{\uparrow \text{mean}}{4}}{\underset{\downarrow \text{standard deviation}}{1}} = 0.25$$

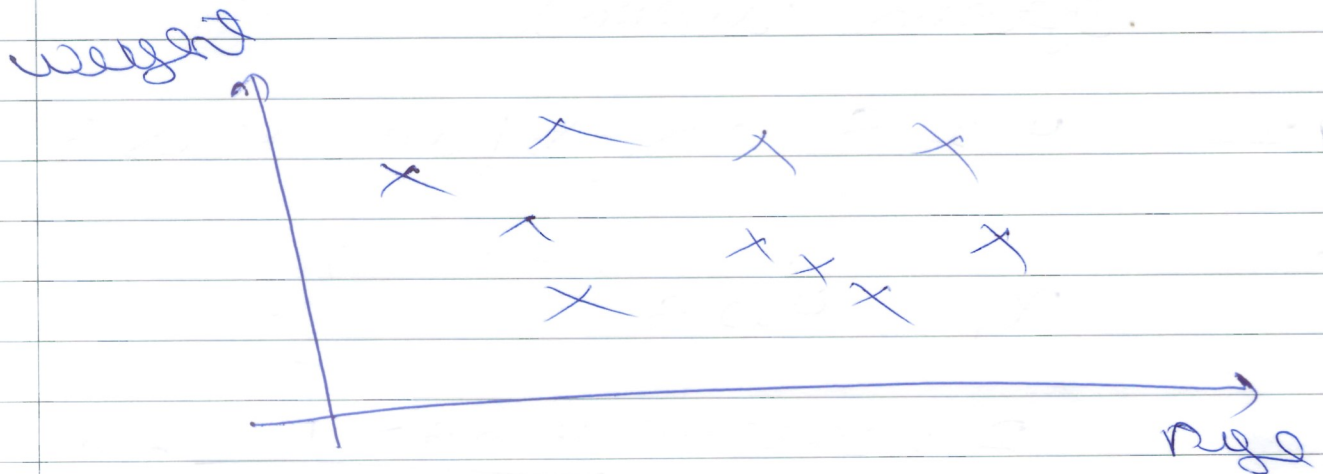
$\therefore$ 4.25 is 0.25 away
from mean.

$$x_i = 2.5$$

$$2.5 - \text{zero} = \frac{2.5 - 4}{1} = -1.5$$

Ex:- Dataset:-

Age	Weight	Height	
Age	Weight	Height	Salary
24	70	175	40K
25	60	160	50K
26	55	180	60K
23	40	130	30K
30	30	175	20K



our machine learning models will optimize themselves in a better way.

(A)

Standardized

$$Z_{score} = \frac{x_i - \mu_{avg}}{\sigma}$$

$$= \frac{x_i - \mu_{avg}}{\sigma}$$

These values like

- (1) Clustering Algorithms
 - (2) Linear Regression
 - (3) Logistic Regression
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